Milestone-2: Data Visualization and Analysis using Power BI

Tools Used

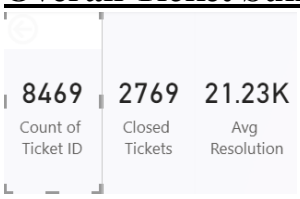
Power BI was used to create interactive dashboards and perform data visualization for IT support ticket analysis.

Exploratory Data Analysis

Exploratory Data Analysis (EDA) was conducted using Power BI to understand patterns in IT support ticket data. Various visualizations were created to analyze ticket types, priorities, resolution times, and distribution across different categories.

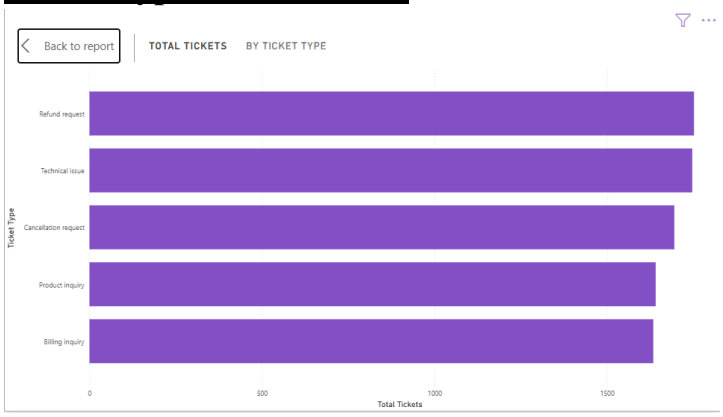
Visualization Insights

1. **Overall Ticket Summary:**



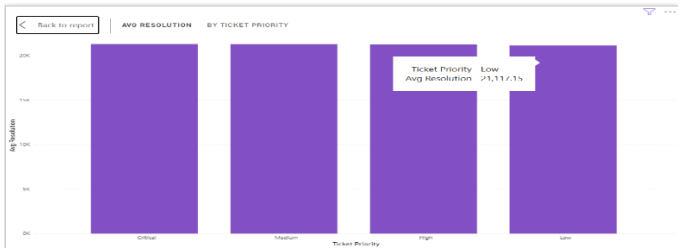
The dashboard shows a total of 8,469 tickets, out of which 2,769 tickets are closed. The average resolution time and response time are approximately 21.23K units, indicating consistent performance across the system.

2. **Ticket Type Distribution:**



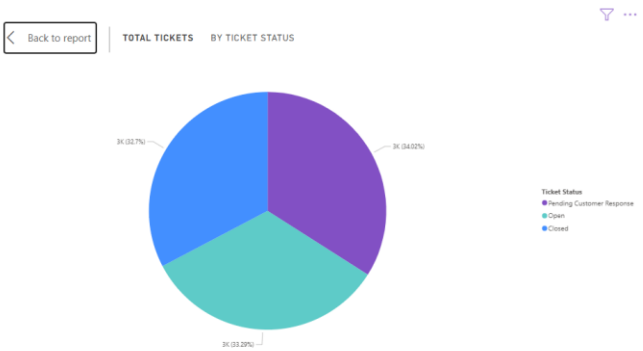
The ticket type distribution shows that technical issues and refund requests contribute significantly to the total number of tickets. This indicates that most user concerns are related to system functionality and transactions.

3. **Priority Analysis:**



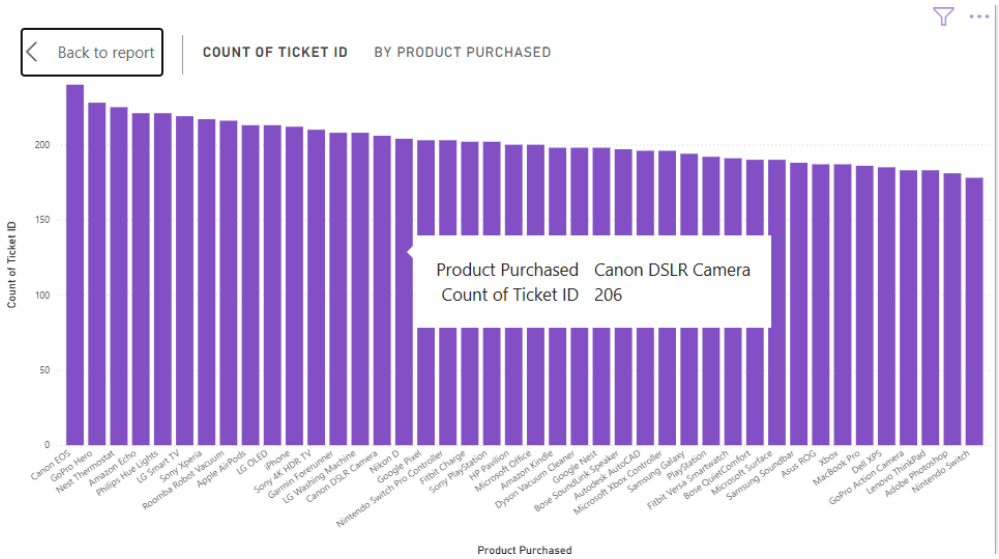
The average resolution time across different priorities (Critical, High, Medium, Low) is almost similar. This suggests that the support team is handling tickets uniformly regardless of priority level.

4. Ticket Status Distribution:



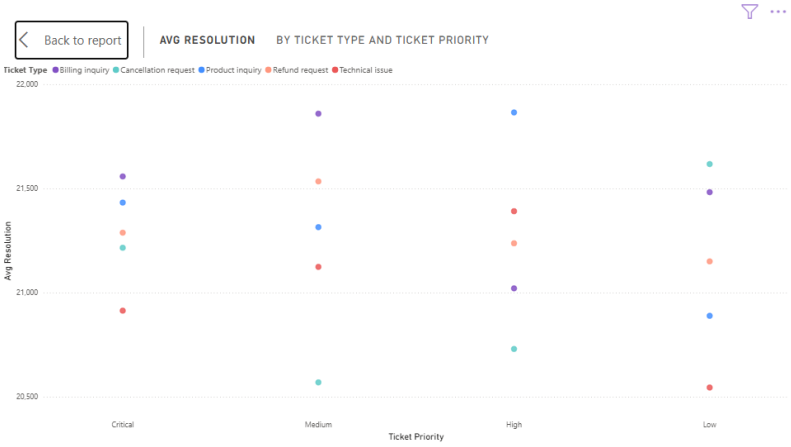
The ticket status chart shows that tickets are almost evenly distributed among Open, Closed, and Pending categories. This indicates a balanced workflow but also highlights that a significant number of tickets are still unresolved.

5.Product-Based Ticket Analysis:



The product-wise ticket distribution shows that certain products generate more support requests. This indicates potential product-specific issues that need attention.

6.Resolution Cluster Analysis:



The comparison between fast and slow resolution clusters shows that most tickets fall under the fast resolution category. However, slow-resolution tickets still exist and require optimization.

Key Insight:

- The majority of tickets are related to technical and transactional issues
- Resolution time is consistent across all priority levels
- A significant number of tickets remain open or pending
- Certain clusters show repeated issues (high similarity)
- Multiple communication channels are actively used
- Some products generate higher ticket volumes
- Most tickets are resolved quickly, but slow-resolution cases need improvement

Conclusion:

In this milestone, Power BI dashboards were used to analyze IT support ticket data. Various patterns and trends were identified, including ticket distribution, resolution efficiency, and cluster behavior. These insights will be used in the next milestone to evaluate performance and optimize support operatisons.